

What are the different layers of a data warehouse?

DATA WAREHOUSING CONCEPTS



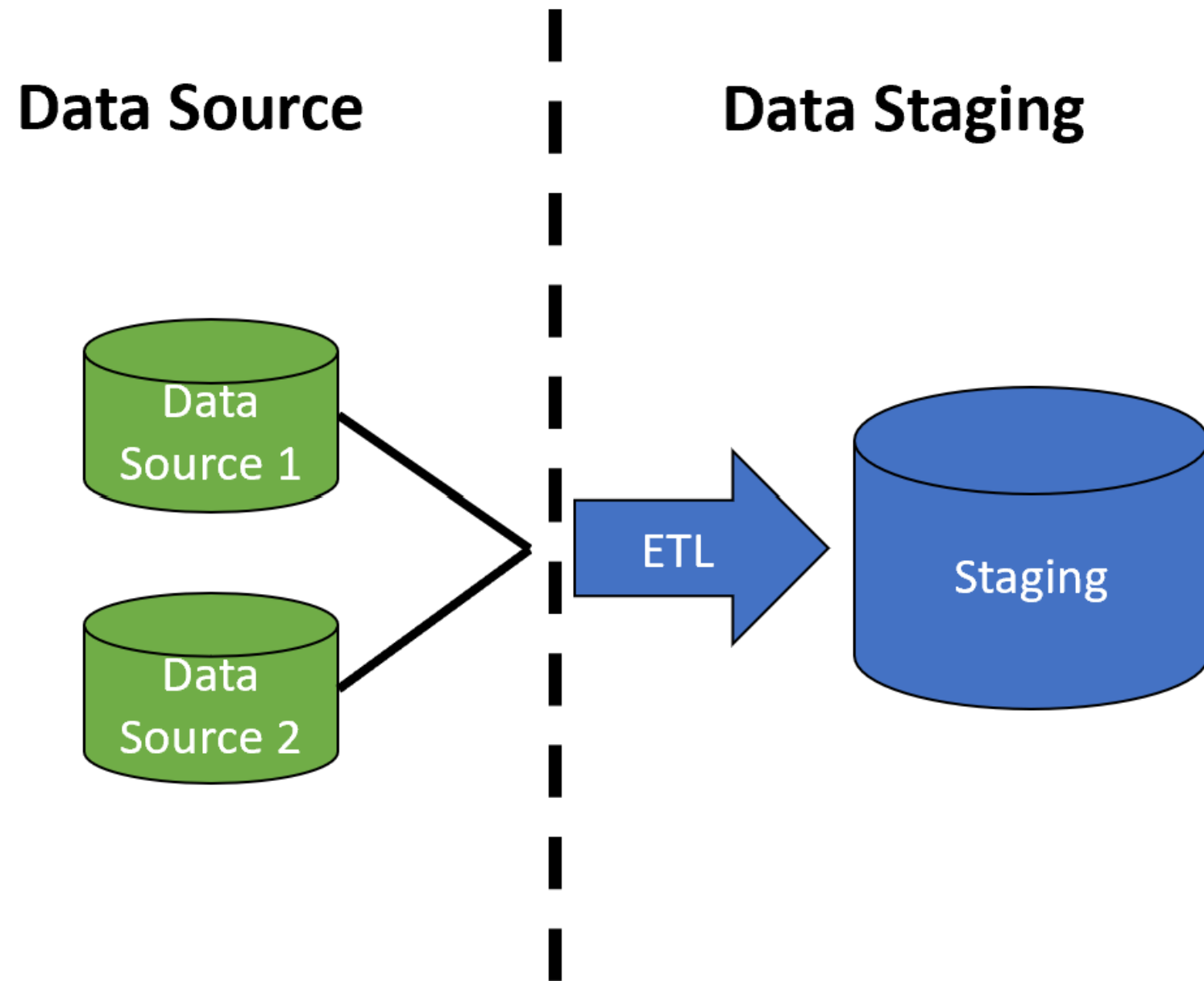
Aaren Stubberfield
Data Scientist

Layer overview - data source

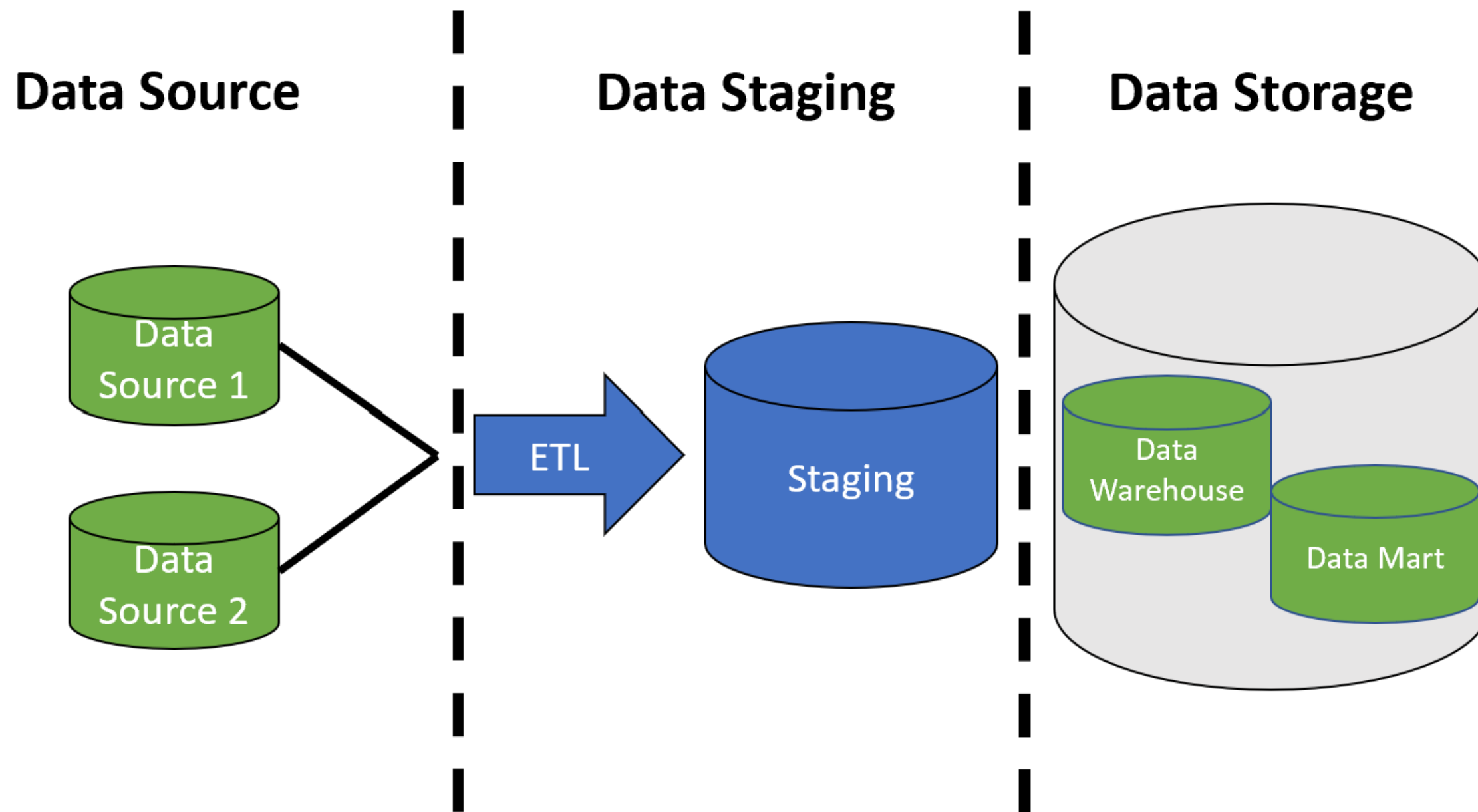
Data Source



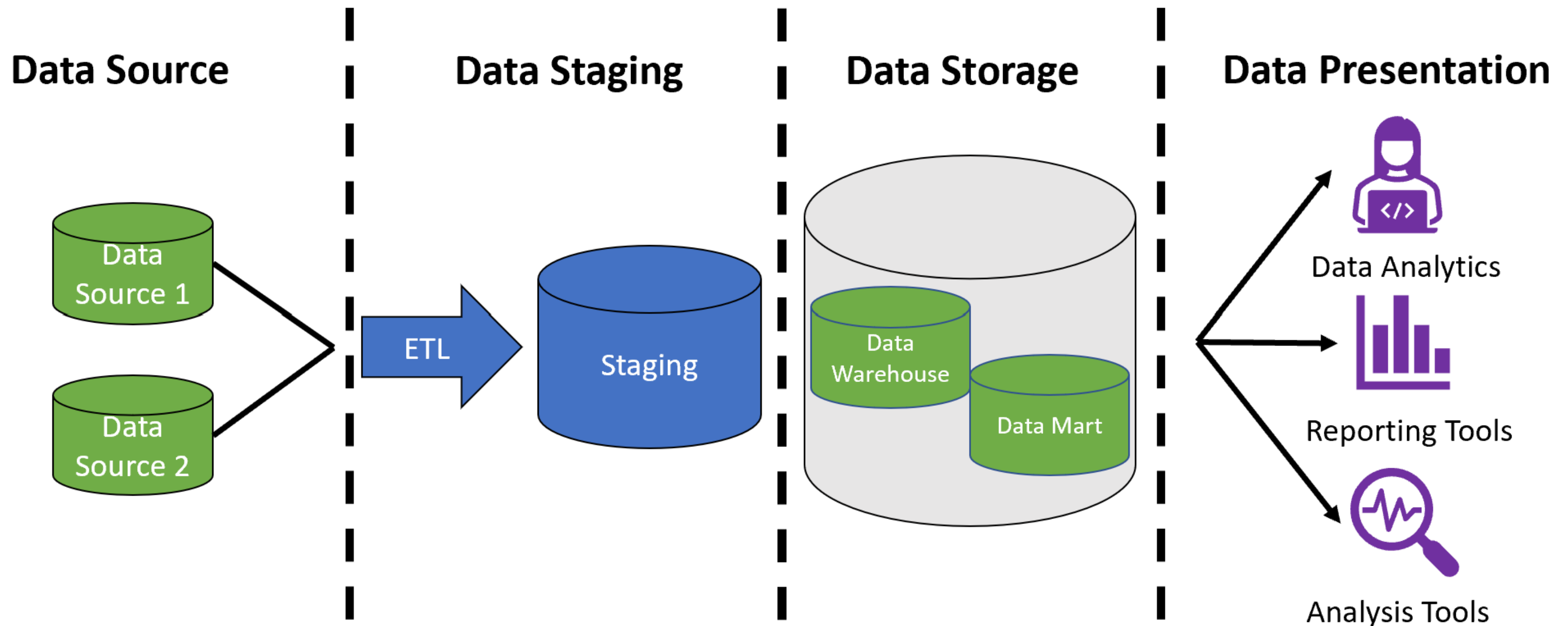
Layer overview - data staging



Layer overview - data storage



Layer overview - data presentation



Data source layer

- All data sources for data warehouse
- Examples of data sources:
 - Transactional database
 - Log files
 - Spreadsheets

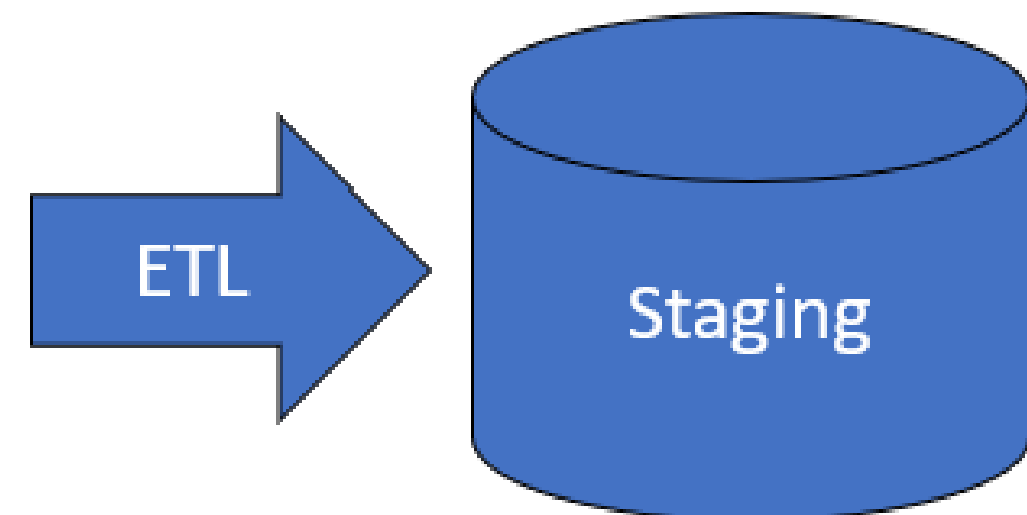
Data Source



Data staging layer

- Layer extracts, transform, and clean data through ETL process
- Contains ETL process and storage tables

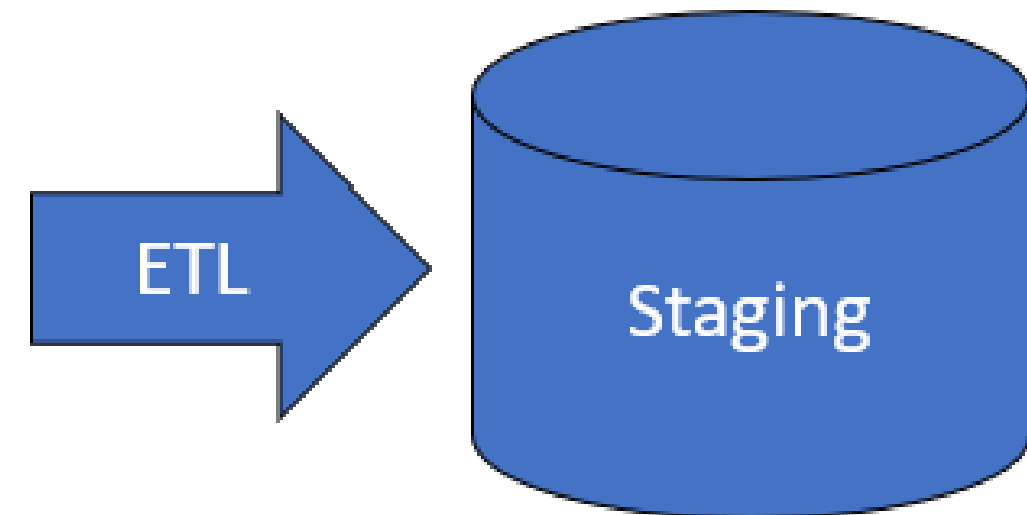
Data Staging



ETL process within data staging layer

- Extracted
- Business rules applied and cleaned
- Staging database often used
- Must be able to extract valid data
- Batch / full loading

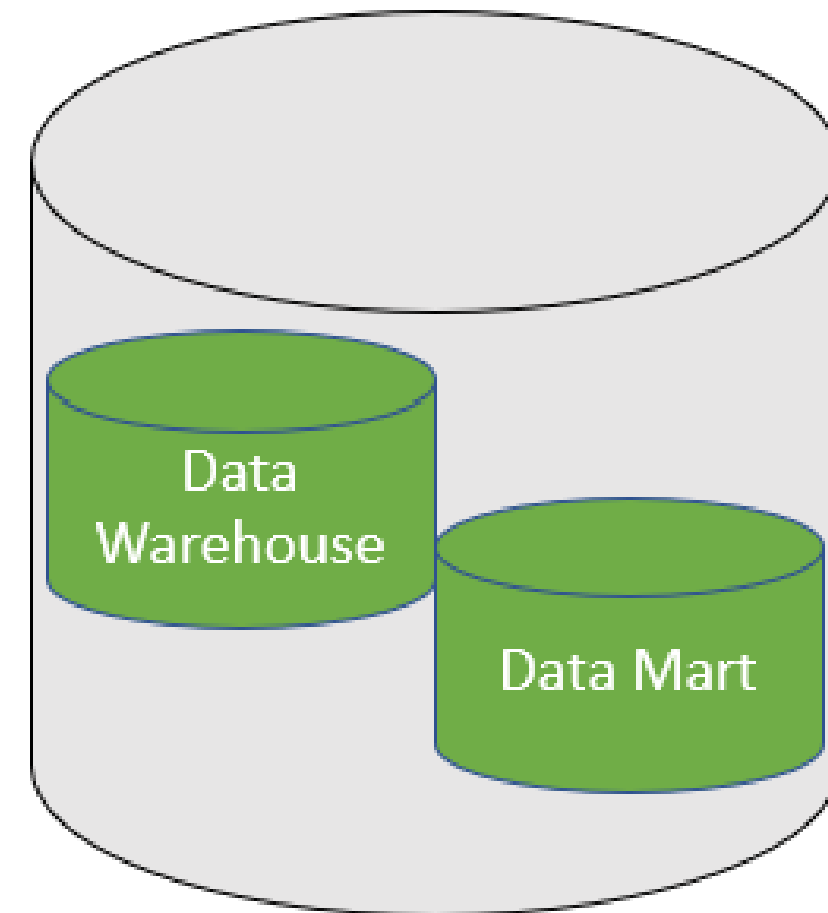
Data Staging



Data storage layer

- Data is stored in warehouse and data marts
 - *Data warehouse -> Data mart*
 - *Data mart -> Data warehouse*

Data Storage



Data presentation layer

- Users interact with stored data
- Users:
 - Use BI (Business Intelligence) tools
 - Use data mining tools
 - Create direct queries

Data Presentation



Data Analytics

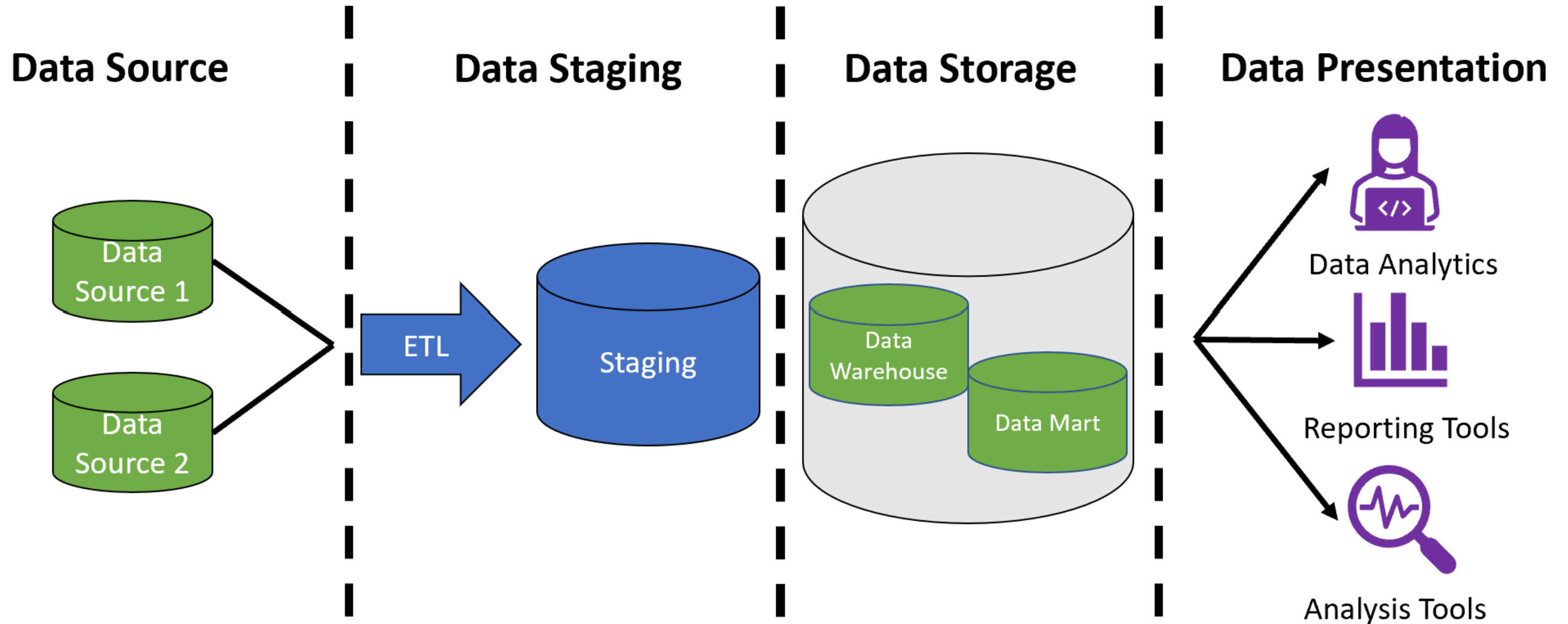


Reporting Tools



Analysis Tools

Summary



Let's practice!
DATA WAREHOUSING CONCEPTS

The presentation layer

DATA WAREHOUSING CONCEPTS



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Presentation layer tools

- Users interact with the presentation layer
 - Area of constant development

Presentation Layer Groups:

- Automated reporting/dashboarding tools
- BI/data analytics
- Direct queries

Data Presentation



Data Analytics



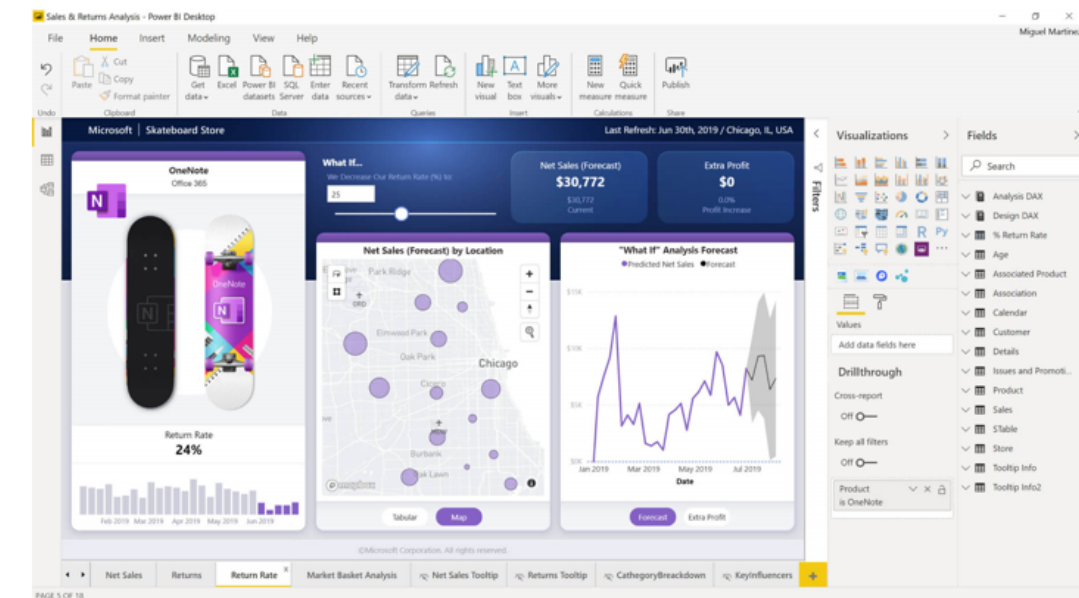
Reporting Tools



Analysis Tools

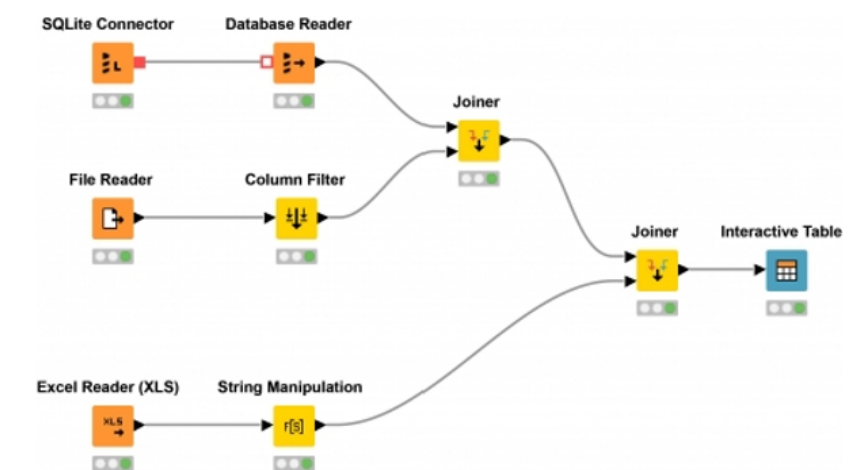
Automated reporting/dashboarding

- Goal:
 - Create reports needed for decision making
 - Create dashboards using historical data
- Users:
 - Analysts
 - Citizen Data Scientist



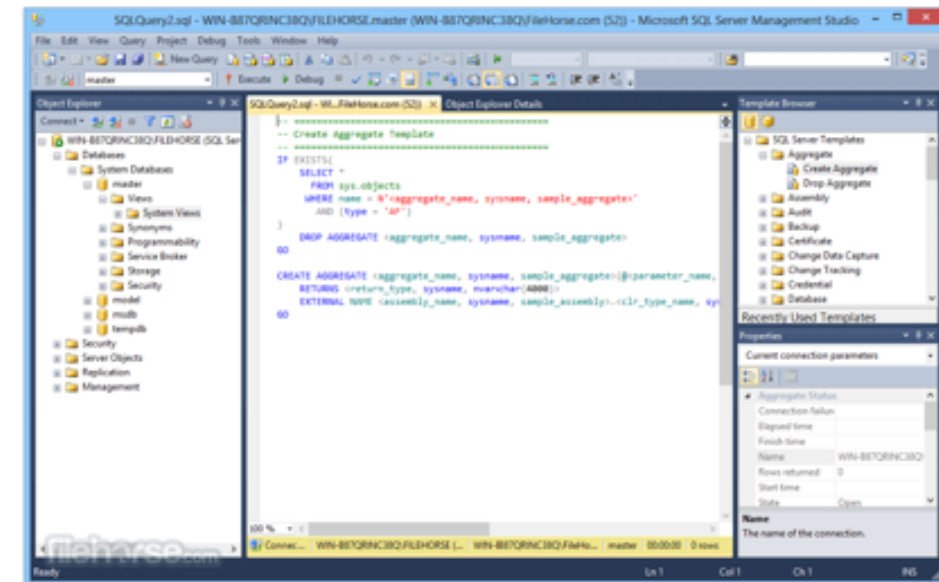
BI/data analytics

- Goal:
 - Tools for exploration
 - Looking for patterns
- Users:
 - Analysts
 - Data Scientist



Direct queries

- Goal:
 - Sophisticated tools for exploration
- Users:
 - Analysts
 - Data Scientist
 - Data Engineer



Let's practice!
DATA WAREHOUSING CONCEPTS

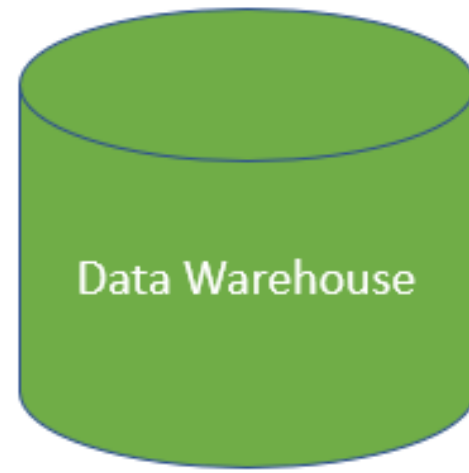
Data warehouse architectures

DATA WAREHOUSING CONCEPTS



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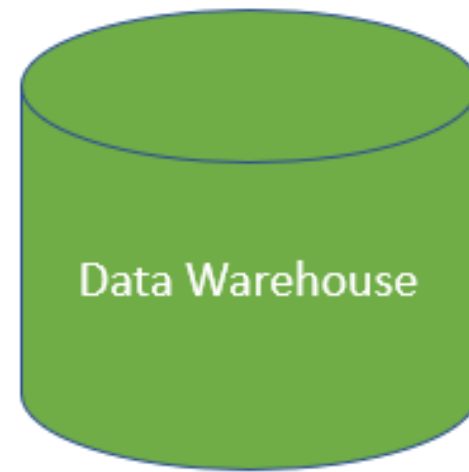
Inmon - top-down



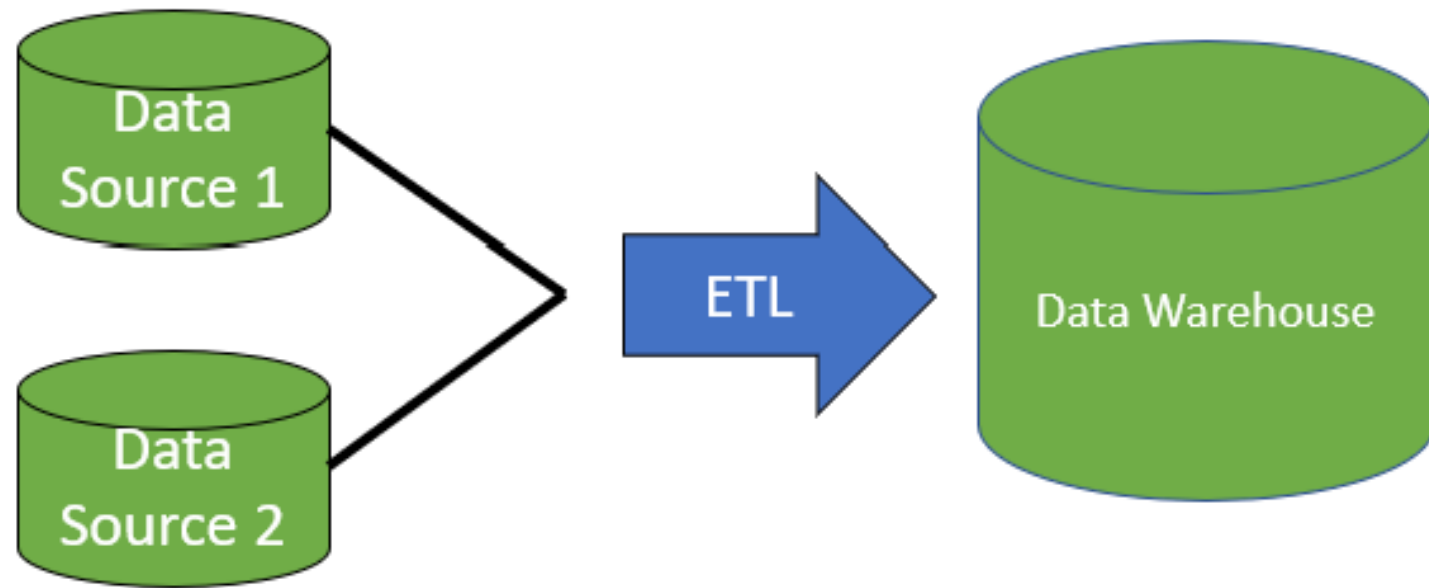
Inmon - top-down

Must decide:

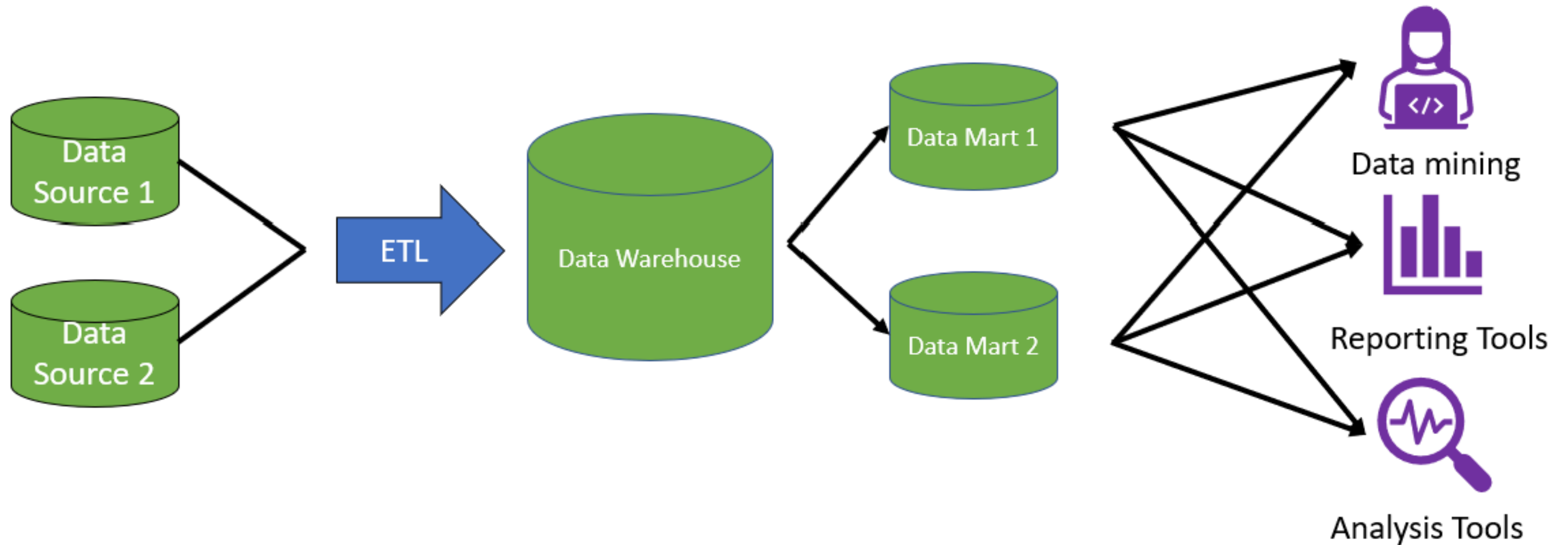
- On all data definitions, cleaning, and business rules
- Before any data enters warehouse



Inmon - top-down



Inmon - top-down



Pros and cons of top-down

Advantages:

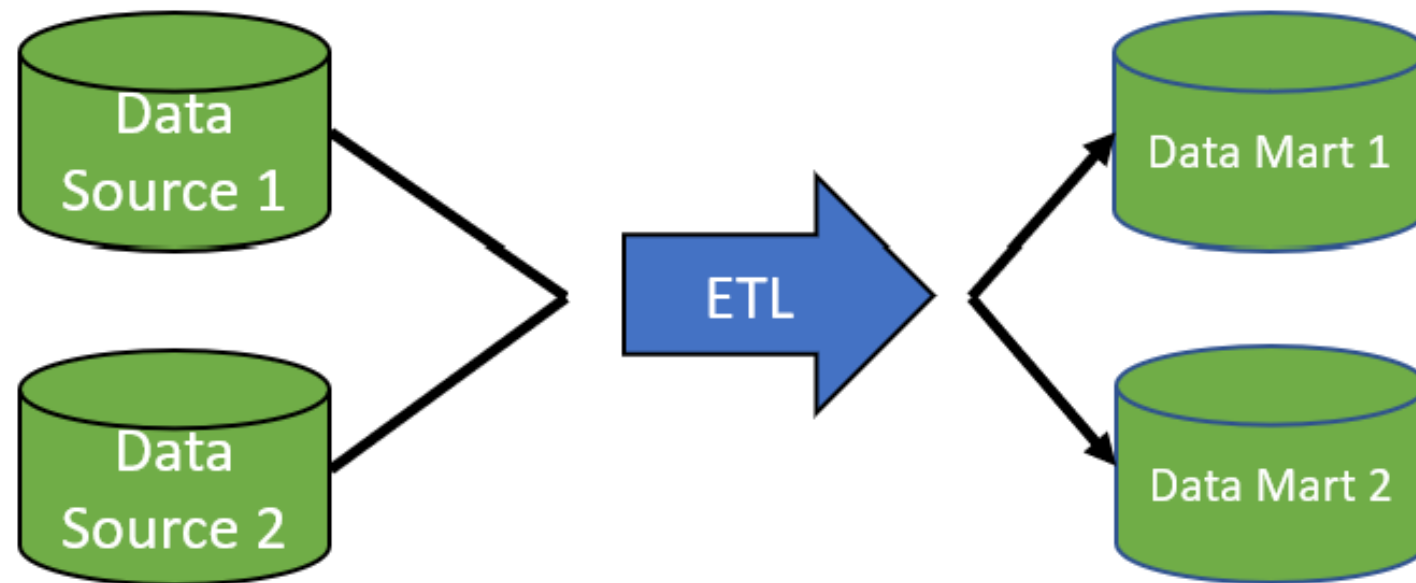
- Single source of truth for organization
- Normalization = less storage
- Easy to change data marts to support reporting changes

Disadvantages:

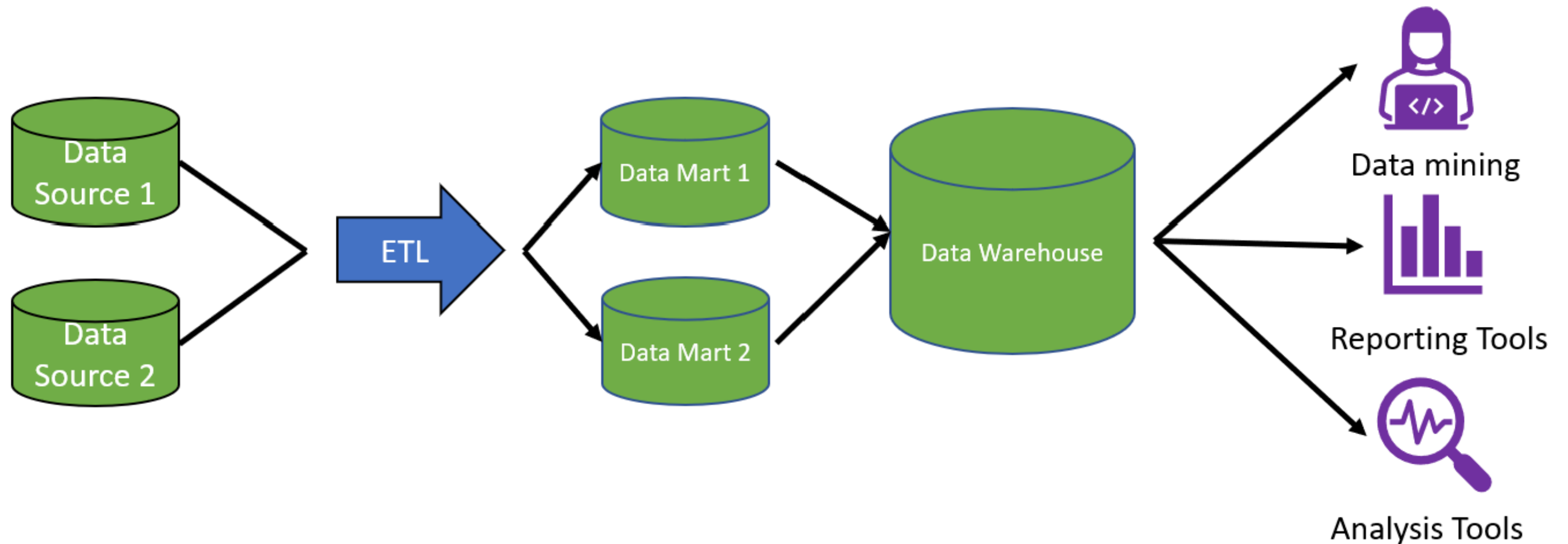
- More joins = slower response time
- Lengthy upfront work
 - Higher startup cost

Kimball - bottom-up

- Denormalizes data
- Focus on departmental data mart
- Data moves directly from ETL to data marts



Kimball - bottom-up



Pros and cons of bottom-up

Advantages:

- Upfront development speed
 - Lower startup cost
- Denormalized = user friendly

Disadvantages:

- Increased ETL processing time
- Greater possibility of duplicate data
- Ongoing development needed

Let's practice!
DATA WAREHOUSING CONCEPTS

OLAP and OLTP systems

DATA WAREHOUSING CONCEPTS



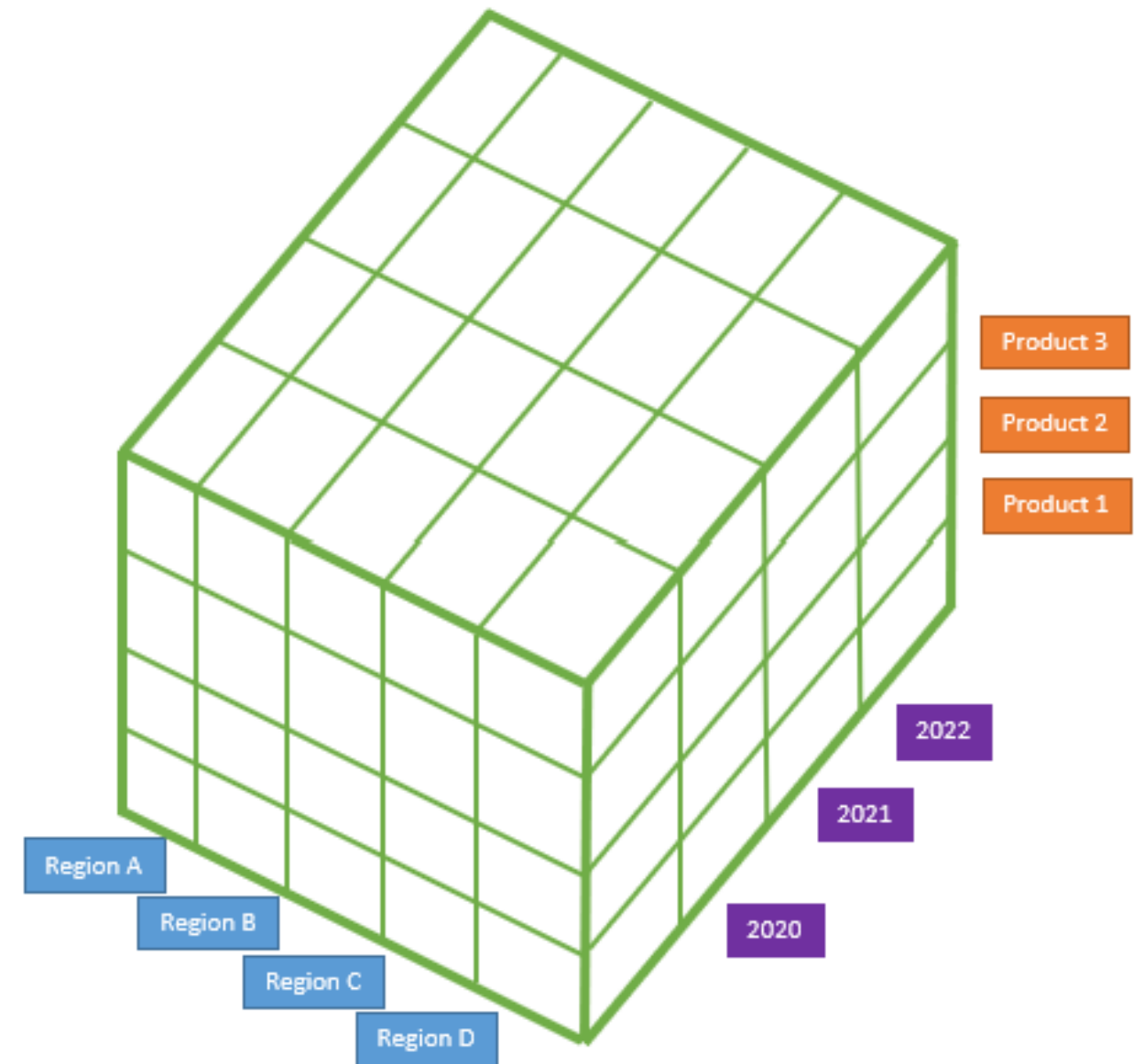
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OLAP systems

- OLAP (*online analytical processing*)
- Designed to support analysis of large amounts of data
- Example dimensional organization:
 - country, state, city
 - years, months, days
- OLAP reorganizes data into multidimensional format

OLAP cube

- OLAP cube key to OLAP system
- Faster processing vs. traditional relational databases
- Hypercubes have more than three dimensions



OLTP

- OLTP (*online transaction processing*)
- Designed for processing simple database queries
- Used in source systems to data warehouse



¹ Photo by Rodnae-Productions on Pexels

Example for a credit card company

OLTP:

- System tracks customer's purchase
- Processes large amounts of simple database updates to account balances

OLAP:

- Designed for analyzing purchase data
- Data organized by multiple dimensions



Summary

Differences	OLAP	OLTP
Optimization:	Complex read-only queries for analysis	Simple queries
Data Representation:	Multidimensional	Rows and columns

Let's practice!
DATA WAREHOUSING CONCEPTS